

SALMI, KuwaitWMO: **405700**Lat: **29.100N**Lon: **46.683E**Elev: **950**StdP: **14.20**Time Zone: **3.00 (E03)**Period: **82-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.0	39.7	7.3	8.3	51.4	10.8	9.9	54.1	26.9	55.2	23.9	60.2	8.9	300	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.2	116.6	69.0	114.9	68.4	112.9	67.7	72.5	98.7	70.6	104.1	69.4	105.7	12.4	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	99.9	80.9	62.6	88.1	77.7	60.0	79.9	73.2	36.8	96.2	35.0	104.1	34.0	106.1	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	27.1	23.9	21.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)				25.2	77.8	5.0	2.3	21.6	79.5	18.7	80.9	15.9	82.2	12.3	83.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.5	55.3	59.7	68.3	77.5	89.1	95.8	98.5	98.5	91.6	82.3	66.5	57.6
	DBStd	16.94	7.29	7.06	7.85	6.26	5.75	5.51	5.15	4.88	5.51	5.73	7.12	5.75
	HDD50	53	33	10	0	0	0	0	0	0	0	0	1	9
	HDD65	839	308	174	53	2	0	0	0	0	0	0	66	236
	CDD50	10452	196	280	567	826	1213	1374	1503	1503	1248	1002	495	245
	CDD65	5763	6	24	155	378	748	924	1038	1038	798	537	110	7
	CDH74	92182	94	223	1679	4440	11632	16116	18685	18589	12902	6724	1009	89
	CDH80	64143	23	58	731	2232	7692	11916	14266	14167	8868	3830	350	10
Wind	WSAvg	9.9	8.7	9.9	10.2	10.5	10.2	11.7	12.3	10.9	9.0	8.5	8.3	8.6
Precipitation	PrecAvg	5.6	1.0	0.7	0.8	0.7	0.2	0.0	0.0	0.0	0.0	0.2	1.1	0.9
	PrecMax	10.0	2.6	2.1	3.2	2.5	0.8	0.1	0.2	0.0	0.0	1.0	6.3	3.2
	PrecMin	2.3	0.1	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.0	0.7	0.6	0.7	0.5	0.2	0.0	0.0	0.0	0.0	0.3	1.4	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.5	86.8	99.1	102.5	112.4	116.7	118.6	118.0	114.9	104.9	92.0	81.2
		MCWB	59.8	58.2	59.9	63.6	66.6	68.7	69.4	69.5	69.3	65.3	60.2	57.6
	2%	DB	75.9	80.7	93.3	98.0	109.0	114.7	117.0	116.1	111.1	102.0	87.8	76.2
		MCWB	57.0	56.7	60.2	62.4	66.0	67.9	68.9	69.2	67.5	64.5	59.5	55.7
	5%	DB	71.3	75.7	88.1	94.5	106.1	112.6	115.1	114.5	108.1	99.2	84.1	72.2
		MCWB	55.0	54.8	58.9	61.3	65.1	67.4	68.4	68.5	66.1	63.7	58.0	55.1
	10%	DB	67.3	71.8	83.2	90.9	103.1	110.2	112.7	112.3	105.4	96.1	80.2	68.5
		MCWB	53.1	53.5	57.4	60.7	64.4	66.2	67.7	67.8	65.2	62.7	57.1	54.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.3	63.3	67.3	67.5	69.9	70.8	74.7	75.1	75.4	74.7	64.3	64.4
		MCDB	74.5	74.3	82.1	87.0	97.5	108.9	98.7	93.9	90.7	87.4	75.1	68.9
	2%	WB	60.0	60.8	64.2	65.7	68.2	69.1	71.1	72.0	70.7	70.0	62.0	61.6
		MCDB	71.0	70.5	79.6	85.8	96.0	111.2	108.0	103.0	103.2	82.4	76.0	69.1
	5%	WB	57.5	58.6	62.0	64.5	66.9	68.0	69.7	70.5	68.2	67.9	60.7	59.0
		MCDB	68.7	70.0	77.8	83.9	97.3	109.6	111.1	106.5	104.9	83.3	75.4	67.9
	10%	WB	54.4	55.7	59.9	63.3	65.7	66.9	68.5	69.4	66.3	65.8	59.3	56.6
		MCDB	65.7	68.5	78.7	82.8	97.8	108.2	109.8	107.6	103.3	87.1	75.6	65.1
Mean Daily Temperature Range	5% DB	MDBR	20.5	21.2	23.2	22.8	23.8	25.0	25.2	25.6	26.3	24.8	21.4	20.8
		MCDBR	25.0	25.3	29.4	26.5	26.6	26.3	26.8	27.0	27.7	27.1	25.8	25.0
		MCWBR	12.1	10.7	11.6	9.5	9.2	10.1	9.8	9.8	10.6	10.5	11.0	12.3
	5% WB	MCDBR	22.6	20.8	22.4	21.6	23.9	25.5	26.2	25.5	26.7	23.1	20.7	20.0
		MCWBR	12.1	10.6	9.8	8.0	9.0	10.4	10.8	10.3	11.4	9.5	9.9	10.8
Clear-Sky Solar Irradiance	taub	0.473	0.523	0.584	0.676	0.689	0.636	0.635	0.595	0.552	0.555	0.526	0.479	
	taud	1.804	1.699	1.577	1.462	1.450	1.512	1.535	1.606	1.680	1.675	1.724	1.802	
	Ebn at Noon	230	231	227	213	211	222	221	228	233	219	212	220	
	Edh at Noon	58	70	84	97	99	92	90	83	75	70	62	56	
All-Sky Solar Radiation	RadAvg	1143	1421	1766	1940	2202	2470	2378	2253	2005	1591	1181	1047	
	RadStd	78	79	72	84	99	83	64	49	62	69	92	87	

Historical Trends

Station Only	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	+0.63	N/A	N/A	+0.94	N/A	N/A	N/A	N/A	+232	+245